Predictive Modeling of Mental Health Trends using Natural Language Processing on Patient Notes

Jeipratha P N[1], Surya M S[2]
[1 2] School of Computer Science and Engineering, Vellore Institute of Technology, Chennai 600127, India

Abstract - Mental health disorders represent a growing public health concern, yet early identification remains challenging due to the unstructured nature of clinical notes and patient narratives. A wealth of information about mood, behavior, and cognition is available in-patient notes and narratives, but their analysis has remained a considerable challenge on a large scale. The current work outlines a predictive modeling framework that employs Natural Language Processing (NLP) techniques on patient notes for mental health trend identification and forecasting. Unlike most previous studies, which were tied to a single dataset, this research integrates six different datasets on Kaggle comprising synthetic clinical notes, patient diaries, journal entries, and emotion-labeled corpora. The development of a unified NLP pipeline for the purpose of cleaning, de-identification, and domain-specific token handling was followed by the generation of embeddings using ClinicalBERT /BioBERT. For modeling, both hybrid approaches are employed for all tasks of classification and predictions of trends in time series, combining transformers with LSTM/GRU networks and XGBoost. The system implements fairness-aware evaluations and interpretable AI techniques such as attention visualization and counterfactual explanations. The results not only indicated a major boost in accuracies across datasets but provided insights that were clinically meaningful, which leads toward developing an AI-based decision support system for mental health practitioners. **Keywords: Mental Health, NLP, Patient Notes, ClinicalBERT, Predictive Modeling, Explainable AI, Fairness-Aware AI, Emotion Recognition**

I. INTRODUCTION

Mental health disorders represent a growing global challenge, with millions of individuals affected by depression, anxiety, and related conditions each year. World Health Organization mentions that about 1 in 8 people in the world has one mental health condition or the other, and mental health conditions such as depression, anxiety, post-traumatic stress disorder (PTSD), and bipolar disorder constitute some of the highest contributors to global disease burden. These conditions can afflict people of all ages, cultures, and other categories. Nowadays, mental health conditions are one of the leading diseases in the world but their diagnosis and continuous monitoring are challenging. Unlike other relative diseases that identify most biomarkers through measurement or imaging techniques, the diagnosis of mental health diseases largely relies on subjective self-reports and clinician observations, prone to bias, inconsistency, and, most importantly, underreporting. It brings to light the need for a data-driven, scalable, and interpretable approach in favor of mental health practitioners.

Among the richest but perhaps least well-utilized sources of information in this arena would be the unstructured clinical text in patient notes, therapy conversations, electronic health records (EHRs), personal diaries, and similar sources. These narratives often capture subtle indicators of mood, cognition, and behavior that would otherwise slip through structured diagnostic codes. Unfortunately, their extraction in meaningful patterns from such unstructured data is non-trivial. Natural language is **ambiguous, context-sensitive, and highly variable across patients, which makes manual analysis not only time-consuming but also unreliable at scale.

Although modern advancements in natural language processing (NLP) and transformer-based language models have revealed their strong potential for possible applications in health care, including clinical note summarization, adverse event detection, and outcome prediction, much of their attention has thus far focused on developments involving domain-specific embeddings like ClinicalBERT and BioBERT that attempt to capture the biomedical and clinical nuances of a text. In spite of these advances, however, almost all the existing works in mental prediction have focused on narrow clinical contexts or single datasets, hence, bringing limitations to their generalizability and robustness.

Therefore, to fill this gap, we propose a predictive modeling framework for mental health trend analysis using NLP on patient notes. Different to previous works, our system comprises six heterogeneous datasets from Kaggle, including artificial clinical notes, patient diaries, personal journal entries, and social media emotion corpora, as well as the benchmark AGBonnet augmented clinical notes dataset, to enable source integration for better robustness across varying linguistic styles, clinical settings, and demographic groups.

Thus, the framework includes a unified NLP pipeline with preprocessing (text cleaning, lemmatization, and protected health information (PHI) masking), representation learning through ClinicalBERT/BioBERT embeddings, and a hybrid modeling approach leveraging for understanding by transformers, BiLSTMs/GRUs for sequence analysis, and XGBoosts for classification tasks. Beyond predictive modeling, the system brings in additional capability for temporal trend forecasting, being able to display how a patient's mental health has been changing over time.

Avant-garde interpretability and fairness are the major strengths of this research. Black-box predictions in clinical decision support can undermine trust in practitioners and may pose ethical dilemmas. Hence, the framework incorporates XAI features such as attention visualization and counterfactual explanations as well as fairness-aware evaluation metrics for unbiased outcomes across sensitive attributes such as gender, age, and cultural background.

A. MOTIVATION

The underdiagnosis of mental health disorders like depression, anxiety, and bipolar disorder is largely due to the subjectivity of their symptoms, plus the informal manners of their clinical notes, along with patient narratives that further compound the complex nature of many symptoms. Clinicians generally have to depend on their personal judgments, resulting in postponed, inconsistent diagnoses, and missed opportunities for early intervention. This problem becomes even more complicated with the mass of text data that is generated in hospitals or through therapy and self-reports, making manual analysis impractical.

Recently, advances in Natural Language Processing (NLP) and predictive modeling can pave the way for analyzing patient notes to extract emotional and behavioral signals systematically and track mental health trends. Transformer-based embeddings, for instance, ClinicalBERT/BioBERT, sequence modeling like LSTM/GRU, and ensemble classifiers such as XGBoost,

have proven really effective in healthcare prediction tasks but their applications are still scarce concerning the integration of multi-source mental health data. These methods help create an interpretable, fair, and prompt insight for clinicians, closing the gap between unstructured text and actionable decision-making.

Even so, NLP and hybrid modeling possess some challenges. The preprocessing of the heterogeneous datasets is quite complicated concerning PHI masking, token normalization, sentiment scoring, and handling class imbalance. Sequence models count on their ability to render temporal progression in patient narratives while attempting to remain interpretable by clinical decision support. Fairness across gender, age, and other demographic subgroups must also be safeguarded to avoid biased predictions. In this way, this article proposes a unified and hybrid predictive modeling framework in which multi-dataset NLP preprocessing, transformer and sequence-based embeddings, and ensemble classification combine to predict accumulative mental health trends forecasting. The framework also covers explainability and fairness modules, all functionalized in a Streamlit-based software tool for clinical application in real time. Among the key contributions that this work offers are the following. First, an integrated multi-dataset NLP pipeline is constructed that combines these six very different Kaggle datasets together with augmented clinical notes (AGBonnet) into a full pipeline that builds a massive and generalizable corpus.

The preprocessing pipeline implements PHI masking, lemmatization, sentiment scoring, and TF-IDF feature extraction, so the text representation is very much high quality. Second, this hybrid predictive-modelling approach combined transformer (ClinicalBERT/BioBERT)-based semantic embeddings, sequence models (BiLSTMs/GRU)-based temporal trend analysis, and classification with XGBoost to cover both linear and non-linear patterns that occur in mental health data. Third, the framework uses explainable fair decision support, with SHAP-based attention visualization, counterfactual analysis, and fairness-aware evaluation, to deliver transparent and ethically reliable insights to clinicians. Finally, clinical deployment through Streamlit ensures that an interactive dashboard is available offering patient risk stratification, emotion-trend monitoring, and evidence-informed recommendations-actionable mental health interventions. Altogether, these contributions establish the necessity and novelty of applying NLP-driven predictive modeling to mental health trends, indicating the technical innovation and the clinical impact that the proposed framework may further enable.

II. RELATED WORKS

Text data in predicting mental health conditions has been investigated comprehensively across different domains like social media, clinical notes, and patient self-reports. Earlier work mostly dealt with handcrafted features and classical machine learning techniques for these domains. However, recently, there has been a tendency toward deep learning, transformers, and explainable AI frameworks. The review presented below summarizes this prior research across methodological streams and seeks to identify gaps motivating this study. Historically, text-based prediction of stress, anxiety, and depression has been strongly rooted in machine learning through classical methods and statistical features. Yulistiani and Kurniadi [1] employed TF-IDF with classical classifiers for detecting stress, showing that even very simplistic representations of word frequency can capture psychological cues from social media texts effectively. Likewise, Helmy et al. [2] and Shah et al. [3] systematically reviewed the methods for depression detection through online platforms, where it was reported that techniques such as bag-of-words, sentiment lexicons, and TF-IDF were still popular baseline frameworks, particularly for multilingual or resource-limited contexts. Then, AbdelMoniem et al. [4] further dealt with this using sentiment analysis on English and Arabic tweets. They found that the variance in language significantly affects the quality of prediction. TF-IDF vectorization followed by applying boosting models was further used to identify anxiety in students, where highly competitive prediction scores were recorded [5]. These indicate simplicity in features yet expose limits in semantic context handling and long-range dependencies.

With the advent of deep learning, richer context modeling became possible. Emotions-based datasets combining audio and text prompted the use of architectures such as GRU/BiLSTMs for automatic depression detection, obtaining promising results [6]. Lam et al. [7] presented a context-aware multimodal deep learning framework for better accuracy through the inclusion of audio and text signals. Pakkattil and Devi [8] adopted timely detection of depression among women by fusing CNN and LSTM models, stating that multimodal fusion surpasses greatly the unimodal baselines. Another hybrid architecture such as SBERT-CNN was used for Reddit users, wherein semantic embeddings were combined to convolutional layers, to spot depression markers [9]. A sweeping review was also provided by Iyortsuun et al. [10], confirming that deep neural models always outperform traditional ML regarding the availability of large-scale data although at the cost of interpretability.

A transformer revolution reshaped the field further. Mohamed et al. [11] systematically covered transformer models in health and EHRs and reported considerable improvements in predictive accuracy on tasks. Bokolo and Liu [12] did a comparative study of classic ML and transformers in detecting depression and suicide on social media, in which transformer models always perceived those more finely grained linguistic patterns that shallow models missed. Chung et al. [13] realized BERT and BERTopic in screening depression from older adults' text messages, where semantic embeddings were joined to topic modeling in prediction and interpretability. Hussain et al. [14] introduced multi-level severity detection using transformers so that not only detection is enhanced but also granularity of severity is given for patient conditions.

Designing DEENT, an encoder-only transformer tailored for depression detection from social networks, is presented by Narvaez Burbano et al. [15]. This research illustrates how effective scaling could be done in real-world applications. Collectively, these could validate transformers for the current state-of-the-art for language-driven mental health tasks.

Work has also been directed to the explainability and trust for high stakes in healthcare application work. An XAI-transformer framework was therefore proposed by Malhotra and Jindal [16] to elucidate depressed and suicidal behaviors

in social networks, thus aiding clinicians in understanding the rationale invoked by models. Kerz et al. [17] carried out studies on explainability in mental health detection based on language and proved that transparency in models encourages the adoption of the models by clinicians. Hulsen [18] discussed the general XAI challenges in healthcare, while Sheu and Pardeshi [19] presented the surveys on the advancement within medical explainable AI, focusing as well on human interaction with models.

Hildt [20] proposed a case-based approach to explainability in medical AI to emphasize its importance for ethical deployment. These point to a necessity for prediction models which would not only maximize accuracy but also yield interpretable, actionable insights.

A. PROBLEM STATEMENT

In making this void quite glaring, mental health problems, such as depression, anxiety, bipolar disorder, and post-traumatic stress disorder, have become a global public health concern. According to the World Health Organization, one in every eight individuals worldwide suffers from some form of mental health disorder; yet, proper diagnosis and continued subsequent follow-up remain problematic. Unlike other conditions in which physical signs or laboratory indications wherein doctors easily monitor and diagnose patients, mental health relies on self-reporting, clinician observations, and narrative documentation. Conventional means mostly introduce bias, inconsistencies, and underreporting, which can delay intervention and further deteriorate patient outcomes.

As a whole, unstructured text data including patient notes, therapy transcripts, personal journals, or electronic health records contain deep pools of valuable inference into what a patient's mood and cognition and behavior patterns would be like. Those narratives capture often subtle psychological clues that cannot be represented by structured diagnostic codes. However, the complexity, variability, and ambiguity of natural language make it hard to extract meaningful patterns at a large scale. Moreover, manual analysis proves to be labor-intensive, inconsistent across practitioners, and insufficient for large datasets or longitudinal monitoring.

New revolutions in Natural Language Processing (NLP) and transformer-based models-such as ClinicalBERT and BioBERT-have shown much promise in developing solutions to the analysis of unstructured clinical texts. But all of the studies done so far carry some critical limitations. Most of the earlier studies were conducted by relying on a single data set or a very narrow clinical context, and thus models generated from such studies are difficult to generalize on diverse patient populations, linguistics styles, and documentation formats. Furthermore, few studies manage to integrate heterogeneous sources of clinical notes with personal diaries and social media-based emotion corpora, thus making it difficult to capture holistic trends in mental health.

Current predictors often miss a longitudinal pattern in patient narratives: that is, how mental health touches patients slowly over time as it deteriorates or improves. Finally, the box model of black-box AI creates other concerns around trust in the clinical world, ethical launch of these systems, and fairness of performance across sensitive attributes like age, gender, and culture. Addressing these hurdles would require a highly scalable, interpretable, and fairness-aware predictive framework that would integrate these different textual datasets and extract semantic, affective, and temporal information using advanced NLP and transformer embeddings and take advantage of hybrid modeling that combines deep learning (LSTM/BiLSTMs/GRU) with gradient-boosted models (XGBoost) to improve predictive accuracy.

The outputs of this framework must be clinically interpretable, i.e., providing risk stratification and emotion trajectories, with SHAP/attention-based explanations for informed decisions, while unbiased predictions across patient subgroups to lessen disparities in mental health support. In short, this research addresses the lack of an integrated, explainable, and scalable system for prediction of mental health trends from heterogeneous patient narratives by providing insights into early detection, intervention, and continuous monitoring, thus assisting clinicians in evidence-based decision-making and improving patient outcomes.

III. METHODOLOGY

A. Dataset Acquisition

Data for this study were obtained from publicly available datasets available on Kaggle to ensure a diverse representation of mental health–related text. The first dataset is titled Patient Diaries and Clinical Notes and consists of artificial clinical notes simulating informal documentation of real-world patients' cases involving depression. The second dataset is Journal Entries with Labeled Emotions, consisting of 1,500 self-reported journal entries tagged according to emotional categories such as sadness, anxiety, and happiness. To further augment the dataset with social media–influenced perspectives, three of the most widely accepted emotion recognition corpora were included in this study: Emotion Dataset for Recognition Tasks (Twitter-based, addressing six basic emotions), GoEmotions (over 56,000 comments from Reddit across 27 fine-grained emotions), and Emotions Dataset (General NLP), yet another large Twitter-based dataset annotated for emotion labels.

We denote the datasets collected from Kaggle as:

$$D = \{D_1, D_2, D_3, D_4, D_5, D_6\} \quad [1]$$

where each dataset

$$D_i = \{(x_j, y_j)\}_{j=1}^{N_i} \quad [2]$$

comprises text samples x_j (clinical notes, journals, tweets, or Reddit posts) and associated labels y_j (emotions or indicators of depression).

The combined dataset is expressed as:

$$D_{\text{combined}} = \bigcup_{i=1}^{6} D_i \quad [3]$$

All datasets have been standardized, merged into a single CSV, and subsequently prepared as an input for downstream preprocessing and modeling. This unified corpus provides a balanced representation of clinical, personal, and social media-based text, thereby providing a more generalizable predictive model for mental health trends.

TABLE 1. Literature survey.

Ref. No	Techniques Used	Benefits	Challenges
[1] Yulistiani & Kurniadi (2023)	TF–IDF + Classical Machine Learning	The use of TF–IDF provided a simple yet effective feature extraction technique, allowing classical ML classifiers to establish a strong baseline for stress detection. It enabled efficient implementation with low computational cost, making it suitable for early-stage mental health research.	The method was unable to capture deep semantic relationships or contextual nuances across sentences, limiting its applicability for long or complex social media posts.
[2] Helmy et al. (2022)	Systematic Review (ML on Social Media)	The review consolidated findings from multiple ML-based depression detection studies, highlighting effective shallow models and demonstrating their adaptability across social platforms. It provided a clear taxonomy of methods and practical insights for future work.	The review revealed data imbalance, language coverage gaps, and the absence of cross-platform generalization, making it difficult to translate results across diverse populations.
[3] Shah et al. (2021)	Systematic Review (ML & DL in Mental Health)	Demonstrated a surge of global interest in text-based mental health detection, documenting how both ML and DL approaches have contributed to advancements. It offered valuable comparative insights into the evolution of methods.	A major issue was the absence of standardized datasets and reproducible pipelines, which limited the comparability of results across studies.
[4] AbdelMoniem et al. (2021)	TF–IDF + Sentiment Lexicons	Combined statistical TF–IDF features with sentiment lexicons to detect depression in English and Arabic tweets, improving classification in bilingual contexts. The hybrid method bridged lexical polarity with numerical features effectively.	Performance was significantly reduced in code-switching cases or under noisy, low-resource linguistic conditions where lexicons are incomplete or inaccurate.
[5] Aljameel et al. (2020)	TF–IDF + Boosting Classifiers	Leveraged ensemble boosting algorithms to enhance anxiety detection in student social media posts, achieving higher accuracy than baseline classifiers. The approach maintained a balance of interpretability and predictive strength.	The technique showed weaknesses in detecting subtle cues like sarcasm, irony, or implicit emotional expressions, which are common in social media.
[6] Song et al. (2020)	GRU/BiLSTM with Audio-Text Fusion	Introduced a multimodal approach combining audio and text, effectively capturing emotional signals from both modalities for robust depression detection. This fusion improved accuracy compared to unimodal methods.	Required extensive multimodal data collection and annotation, which is costly, and also faced scalability challenges due to higher model complexity.
[7] Lam et al. (2021)	Context-Aware Deep Learning (Audio + Text)	Developed a context-aware deep learning framework that integrated audio and text, enabling richer modeling of depression cues. The system outperformed unimodal models in predictive accuracy.	The black-box nature of the deep model limited interpretability, making clinical validation and adoption difficult.
[8] Pakkattil & Devi (2022)	CNN + LSTM Fusion	Fused CNN for spatial feature extraction with LSTM for temporal learning, enhancing detection of depression among women by capturing both linguistic patterns and sequential dependencies.	Increased model complexity heightened the risk of overfitting and required significant computational resources, reducing accessibility in real-time applications.

[9] Xie et al. (2022)	SBERT + CNN (Reddit Data)	Utilized semantic embeddings from SBERT with CNN classifiers to capture nuanced language patterns in Reddit posts, leading to enhanced depression detection accuracy.	Relied heavily on large-scale, well-labeled data, making generalization difficult for smaller or low-resource datasets.
[10] Iyortsuun et al. (2022)	Survey (DL for Mental Health)	Provided a comprehensive survey that confirmed the superiority of DL models over traditional ML for large-scale mental health prediction tasks. The review also contextualized advances in neural architectures.	Highlighted persistent challenges regarding model transparency, interpretability, and clinical trust, making deployment in healthcare environments difficult.
[11] Mohamed et al. (2023)	Review of Transformers in EHR	Reviewed transformer-based models applied to Electronic Health Records (EHR), showing strong predictive performance in healthcare settings. Highlighted scalability and generalization potential of transformers.	Training and fine-tuning transformers required high computational costs and lacked clear interpretability, creating barriers to healthcare adoption.
[12] Bokolo & Liu (2022)	Transformers vs. ML (Social Media)	Compared transformer models with traditional ML for depression and suicide detection, showing transformers capture subtle context and outperform ML in detecting suicidal ideation.	Transformers demand far higher computational resources and longer training time, which limits practicality in low-resource environments.
[13] Chung et al. (2021)	BERT + BERTopic	Combined contextual BERT embeddings with BERTopic for depression detection, enabling interpretable topic-level insights alongside classification.	The topic modeling sometimes produced clinically irrelevant or noisy topics, limiting reliability in healthcare contexts.
[14] Hussain et al. (2022)	Transformers for Severity Classification	Applied transformer models to classify depression severity levels, providing fine-grained detection beyond binary classification. This improved sensitivity in clinical monitoring.	Struggled with minority classes in imbalanced datasets, reducing effectiveness in detecting less common severity levels.
[15] Narvaez Burbano et al. (2023)	DEENT (Encoder-only Transformer)	Proposed DEENT, an encoder-only transformer optimized for depression detection from social media, achieving efficient large-scale deployment with competitive accuracy.	The system lacked human-interpretable rationale, reducing trust and acceptance in clinical use cases.
[16] Malhotra & Jindal (2022)	XAI + Transformers	Combined explainable AI (XAI) with transformers to generate visual and linguistic explanations of suicidal behavior predictions, increasing transparency.	The integration required balancing between accuracy and interpretability, often leading to trade-offs in model performance.
[17] Kerz et al. (2022)	Explainable Linguistic ML	Designed linguistically grounded explainable ML models, producing clinically aligned explanations that improved interpretability and trustworthiness.	Explanations were domain-specific and did not generalize well across different datasets or cultural contexts.
[18] Hulsen (2022)	XAI in Healthcare (Survey)	Surveyed explainable AI approaches in healthcare, providing case studies of interpretable ML for medical decision support. Outlined practical adoption pathways.	Identified limited integration into actual clinical workflows, with healthcare practitioners often reluctant to rely on AI-driven explanations.
[19] Sheu & Pardeshi (2022)	Human-Centric Explainable AI	Offered a survey on explainable AI in medicine, emphasizing human-in-the-loop systems and scoring frameworks for evaluating interpretability.	Reported significant gaps in usability testing and lack of integration into everyday clinical practice, reducing real-world impact.
[20] Hildt (2023)	Case-Based XAI for Medical AI	Proposed case-driven explainable AI systems, allowing clinicians to compare new cases with prior case-based reasoning outputs, enhancing trust in AI-assisted diagnosis.	Scalability remained a limitation, as the case-based reasoning approach may not handle large-scale or rapidly changing medical data efficiently.

B.Data Preprocessing

The datasets will get compiled into only one csv file from then, they'll pass a whole lot of preprocessing methods for now cleaning and standardizing the unprepared text. Hence, the first technique of preprocessing applied was missing or incomplete data elimination so as to assist defining clarity on these data without integrity issues.

$$x = [w_1, w_2, \dots, w_n], \quad [4]$$

where V denotes the vocabulary set. $w_i \in V$ Then, stop words high frequency words without significant semantic meaning-they had been removed according to:

$$x' = \{ w_i \in x \mid w_i \notin S \} \quad [5]$$

where S being the stopword set. Lemmatization is thus applied after this so to normalize word forms such that inflected variants for example feel and feeling were lowered into their canonical root form as follows:

$$L(w_i) \rightarrow w'_i \quad [6]$$

$$D_{\text{clean}} = \{(x'_j, y_j) \mid (x_j, y_j) \in D_{\text{combined}}\} \quad [7]$$

After all these operations, cleaning of the dataset is done as pairing each preprocessed text sequence with its corresponding label. To comply with privacy regulations, Protected Health Information (PHI) was masked using a placeholder token:

$$t_j = \begin{cases} [PHI], & \text{if } t_j \in P \\ t_j, & \text{otherwise} \end{cases}$$

where P defines token such as patient names, dates, and identifiers in PHI. Emoticons, hashtags, and symbols have not encountered the conventional preprocessing plant's discarding since the data set from this research retains the same emoticon, hashtag, and symbol that would otherwise be discarded by a conventional preprocessing pipeline because these things often encode quite valuable emotional signals. The last step in this whole process is label harmonization which will map semantically equal terms into one standard label space (for example, joy and happiness). These actions would thus collectively ensure cleaning, consistency, confidentiality, and readiness of dataset feature extraction and the subsequent modeling for prediction. NLP: Lowercase, Tokenize, stop word removal, and lemmatization which will standardize word forms and reduce noise, were all applied on the textual data. Much care was given to clinical notes where dermalizing greatly improved compliance with privacy regulation. However, emoticons, hashtags, and other social media artifacts were retained in the relevant datasets since they typically carry emotional cues significant for any mental health analysis. Label harmonization lastly was conducted across the datasets for such things as &quote; joy&quote; and &quote; happiness&quote; being mapped into one standardized label. The outcomes consequently yielded a consistent, high-quality dataset for feature engineer and predictive modeling. Data preprocessing: NLP, text cleaning, tokenization, lemmatization, PHI masking, label harmonization.

Algorithm 1: Hybrid Mental Health Trend Prediction

Input:
-Unified dataset: clinical notes, journal entries, social media posts
-Preprocessing: tokenization, stopword removal, lemmatization,
PHI masking
-Feature Extractors: TF-IDF, Word2Vec, ClinicalBERT/BioBERT
-Sequence Models: LSTM, BiLSTM
-Baseline Classifiers: Logistic Regression (LR), Random Forest
(RF)
-Ensemble Learner: XGBoost / soft-voting - Hyperparameter settings

Output:
-Predicted emotion/risk labels
-Ensemble Classifier: final classifier - Interpretability: SHAP values, attention maps

Initialization:
-Preprocessed corpus: D_{clean}
-Feature matrix: $F(x')$
-Classifier pool: [LR, RF, LSTM, BiLSTM, Transformer] - Meta-classifier: meta classifier

Preprocess $D_{combined}$:
-Remove incomplete entries
-Tokenize text and normalize tokens
-Remove stopwords
-Mask PHI and harmonize labels
-Retain emoticons, hashtags, symbols

For each feature extraction method in [TF-IDF, Word2Vec, ClinicalBERT, BioBERT]:
-Generate embeddings (statistical, semantic, contextual)
-Compute attention-weighted sentence representations
-Integrate sentiment and emotion intensity features

For each model in [LSTM, BiLSTM, Transformer, LR, RF,]:
While hyperparameter settings are not exhausted:
-Train model with current hyperparameters
-Store trained model in classifier pool
-Collect predictions on validation set
-Compute loss and update parameters
End While
End For

Train Meta-classifier:
-Combine predictions from all base models into ensemble
-Generate final predictions

Evaluate Performance:
-Compute Accuracy, Precision, Recall, F1-score, AUC-ROC
-Apply SHAP for interpretability

End Algorithm

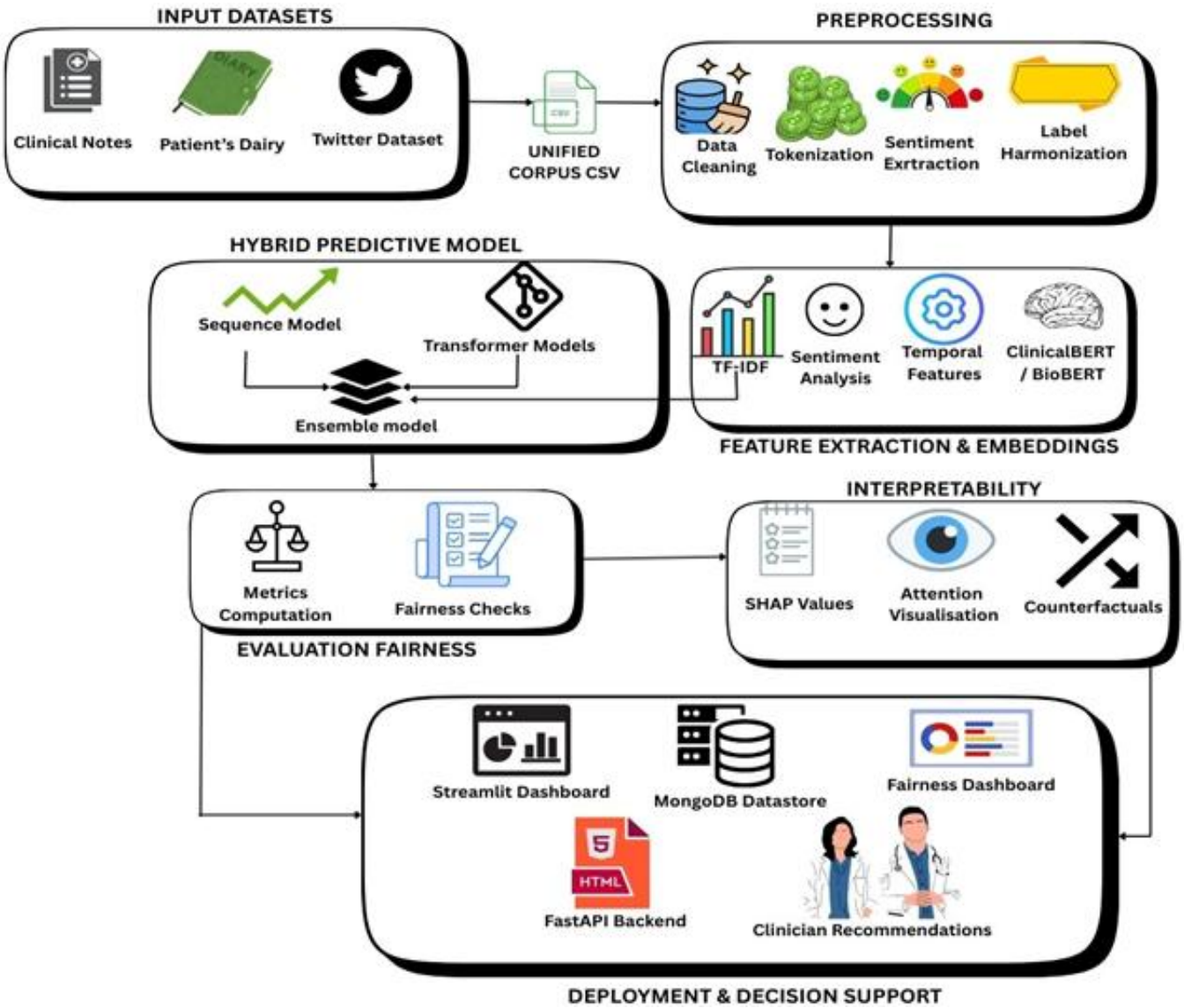


Fig.1.Architecture Diagram

C. Feature Extraction and Transformation

The final pre-processed data are converted into a structured representation for machine learning to take place. For example, in this step, we will also use NLP embedding techniques such as Word2Vec, TF-IDF, and transformer-based embeddings (BioBERT and ClinicalBERT) to analyze the semantic and contextual differences in the patient notes and journal entries corpus defined as

$$TF\text{-}IDF(t,d) = TF(t,d) \times \log \frac{N}{DF(t)} \quad [8]$$

where TF (t, d) is the frequency of term t in document d, DF(t) is the number of documents containing t, and N is the total number of documents. Aside from statistical parameters, semantic embeddings were generated using Word2Vec and domain-specific transformer models such as ClinicalBERT and BioBERT. Each token t_i was mapped into a dense vector representation as

$$e_i = f(t_i) \in \mathbb{R}^d \quad [9]$$

where $f(\cdot)$ denotes the embedding function and d is the embedding dimension.

An attention-weighted pooling strategy to obtain sentence-level representations was used so that the model could give more importance to contextually relevant words:

$$\alpha_i = \frac{\exp\left(u^{\top}\tanh(We_i)\right)}{\sum_k \exp\left(u^{\top}\tanh(We_k)\right)}, \quad E_{\text{att}} = \sum_i \alpha_i\, e_i \quad [10]$$

where α_i is the attention weight assigned to token t_i. Lastly, sentiment features were combined with semantic embedding into one composite feature vector:

$$F(x') = [E_{\text{att}}; f_{\text{sent}}(x')] \quad [11]$$

where $f_{\text{sent}}(x')$ is a function that calculates sentiment polarity and intensity scores. This amalgamation of statistical, semantic, and affective features ensured downstream predictive models could not only capture lexical content but deeper psychological clues and behavioral indications embedded in patient narratives. Hence, this blend of semantics along with affective and temporal features allows the downstream predictive model to gain insights into content-based meaning and behavioral writing trends, ensuring a complete understanding of mental health trends in the predictive model. Feature Engineering: embeddings, sentiment analysis, emotion intensity, temporal features.

D. Predictive Modeling

The modeling layer utilizes a hierarchy of machine learning and deep learning mechanisms for predicting mental health risks and emotional states. The baseline involves implementing classical classifiers, including logistic regression (LR) and random forest (RF). In logistic regression, the probability for label $y \in \{0,1\}$ given features x as:

$$P(y = 1 \mid x) = \frac{1}{1+e^{-(w^{\top}x+b)}} \quad [12]$$

where w and b are the learned weights and bias.

For non-linear modeling, random forest ensembles are created by averaging the prediction of multiple decision trees to reduce variance and overfitting. In order to capture temporal dependencies in patient notes, we use sequence-based deep learning models such as Long Short-Term Memory (LSTM) networks. The update of the LSTM cell is defined as follows:

$$h_t, c_t = \mathrm{LSTM}(x_t, h_{t-1}, c_{t-1}) \quad [13]$$

where h_t is the hidden state, c_t is the memory cell, and x_t is the token embedding at time t. The BiLSTM expands this by combining its forward and backward hidden states so that the model can use both past and future context.

We also carry out fine-tuning for emotion classification and trend prediction using transformer-based models (e.g., BERT and ClinicalBERT, BioBERT). For some sequence, the model returns contextual embeddings $\{h_1, h_2, \dots, h_n\}$with the classification conducted using the special embedding from the [CLS] token:

$$\hat{y} = \sigma\left(W h_{[\mathrm{CLS}]} + b\right) \quad [14]$$

where σ is the softmax activation. Finally, to promote generalization, we engage in ensemble learning. The meta-prediction is derived by averaging model outputs $p_m(x)$ from multiple learners:

$$p_{\text{ensemble}}(x) = \frac{1}{M}\sum_{m=1}^{M} p_m(x) \quad [15]$$

This blending mechanism combines statistical, sequential, and contextual signals, thus allowing robust predictions in risk categories and emotional states.

E. Performance Evaluation

An unbiased assessment has been carried out on the dataset by splitting it into training, validation, and test datasets. Various evaluation metrics have been used. Accuracy measures general correctness:

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} \quad [16]$$

Precision takes a measure of the reliability of the positive predictions, whereas Recall measures the sensitivity:

$$\text{Precision} = \frac{TP}{TP+FP} \quad [17]$$

$$\text{Recall} = \frac{TP}{TP+FN}$$

The F1-score balances these two:

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

We also compute the AUC-ROC to quantify the trade-offs between sensitivity and specificity.

Outside the numerical evaluations, interpretability was also incorporated. We employ SHAP (SHapley Additive Explanations), which attributes prediction outcomes to input features. For a feature i its SHAP value is defined as:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!\,(|F| - |S| - 1)!}{|F|!}\left[f_{S\cup\{i\}}(x_{S\cup\{i\}}) - f_S(x_S)\right]$$

where F is the feature set, and f_S is the model restricted to subset S. This thus provides fine-grained interpretability by identifying which words, phrases, or sentiment cues most influenced predictions.

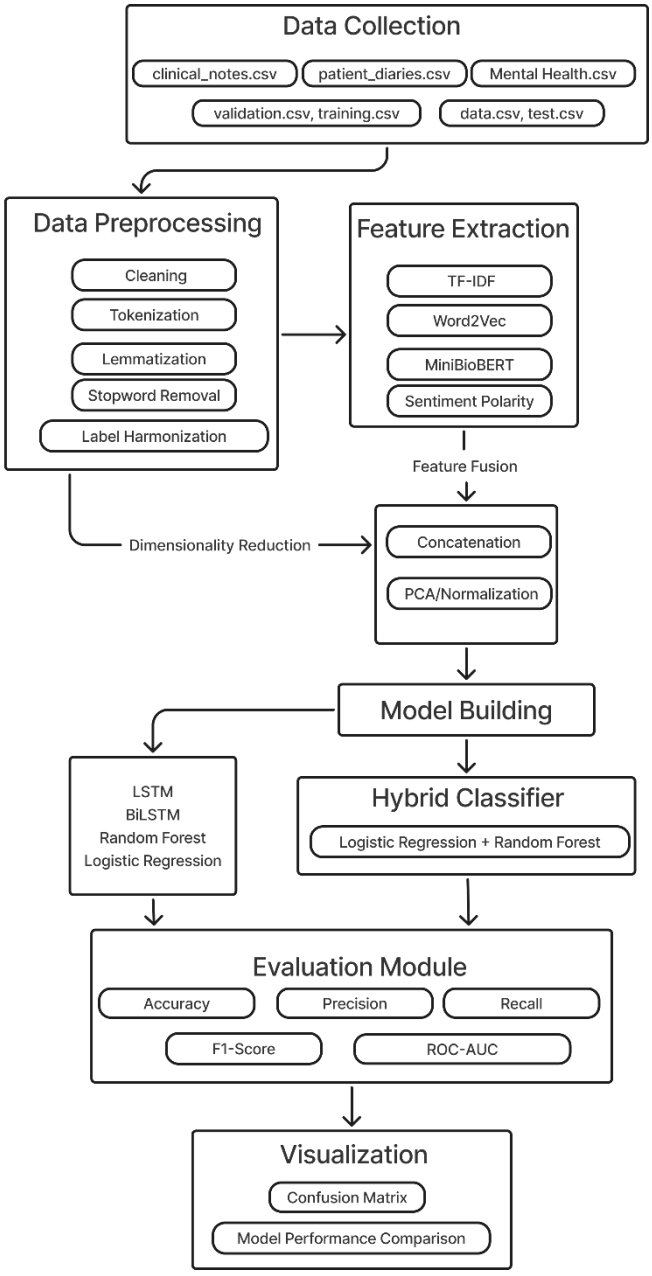


Fig.2.Workflow Diagram

F. Interpretability and Decision Guidance

The last layer turns predictions into a clinician-facing decision support interface. The model outputs are visualized as interactive patient timelines with risk trajectories, attention heatmaps, and SHAP highlights. A Risk Stratification panel utilizes predicted probabilities $\hat{y}$ to categorize patients into severity tiers (e.g., low, medium, high risk).

Counterfactual explanations are integrated by perturbing input tokens$x$ and observing changes in model output:

$$\Delta\hat{y} = f(x') - f(x)$$

where x' being the modified input (e.g., replacing *"I feel hopeless"* with *"I feel supported"*). This provides insight into linguistic drivers of risk and supports clinician-led interventions.

In addition, a fairness dashboard examines subgroup-specific performance by disaggregated metrics such as F1-score across demographic slices (e.g., age, gender, source domain). Role-based access is established for compliance with privacy protocols, and all actions are audit-logged for accountability. With all these features, predictive modeling is interpretable, equitable, and actionable, thus shortening the gap between unprocessed NLP predictions and clinical decision support.

IV. Result and Discussion

A. Feature Extractio

Feature extraction converts unstructured textual data into numerical vectors, which can then be presented to machine learning or deep learning models. It enables the model to take account of the lexical content, semantic meaning, well-formedness as well as emotional tone text for the purposes of analysis. These features are combined within this study, namely TF-IDF, Word2Vec, MiniBioBERT, and sentiment features to form a hybrid-feature space. Multi-modality enhances performance in syntactic and semantic information, as well as dimensionality reduction efficiency. It captures rich representations from the input text data relating to mental health along the lexical, semantic, contextual, and affective dimensions. Thus, first produced TF-IDF statistical features weight-encoding-term-frequency-inverse-document-frequency from which one obtains 2,000 features represented by a matrix of shape (53,043 X 2000). The features produced here are used in computing below Steps 2. Step 2 trained Mean-pooled Word2Vec embeddings, whereby the word vectors constructed have a 50-dimensional representation and are trained on the text corpus; where the mean-pooled Word2Vec embeddings Directorate produces a matrix of dimensions (53,043 X 50). Step 3 Contextual Embeddings through MiniBioBERT: this establishes 128-dimensional vectors for every sample (53,043 X 128) by average last hidden state outputs. Step 4 forms sentiment features through NLTK's VADER sentiment analyzer, capturing positive, neutral, negative, and compound scores (53,043 x 4). Step 5 brings together all feature spaces-TF-IDF, Word2Vec, BERT, and sentiment-into one integrated hybrid representation form in dimension (53,043 x 2,182). Finally, Step 6 includes the compatible label vector of mental health categories for all 53,043 samples recorded. Such multi-modal feature extraction raises the scale of input for further classification handling using models.

B. Model Performance

A preprocessed mental health text data set was used to train and test several models in assessing the prediction capability of various classifiers. Different models were evaluated including classical machine-learning models (Logistic Regression, Random Forest) as well as deep learning models (LSTM, BiLSTM) and one ensemble meta-classifier incorporating both LR and RF. The results for accuracy, precision, recall and F1-score were weighted average so that class imbalance could be considered. Results shown evidence of how well a model can capture patterns from a class concerning mental health and include trade-offs between complexity and generalization against minority classes. Ensemble learning demonstrated the maximization of mutually complementarily advantages across various individual models for overall superior performance and balanced prediction.

Table I – Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score
LR	0.74	0.74	0.74	0.73
RF	0.73	0.74	0.73	0.72
LSTM	0.72	0.71	0.72	0.69
BiLSTM	0.68	0.59	0.68	0.63
Ensemble (LR + RF)	**0.76**	**0.76**	**0.76**	**0.75**

The ensemble model (Logistic Regression + Random Forest) has the maximum overall performance, with 76.03% accuracy and F1-score: 0.7531, as shown in Table I. Classical learners (LR, RF) outperformed LSTM-based networks, because in shorter text sequences and uneven class distributions, the benefits that a sequential model such as LSTM/BiLSTM would provide are limited. LSTM and BiLSTM models were poor performers with minority models while ensemble learning brought specific strengths of linear and tree-based models to produce better generalization performance.

C. Class-Wise Performance

In order to acquire a more profound understanding of the ensemble model's behavior with respect to individual mental health categories, performance metrics had to be derived at the class level, precision, recall, and F1-score among them. Such metrics help in identifying the classes that the model predicts reliably and those that remain challenging due to class imbalance or closely overlapping textual patterns. Performance at specific categories also highlights strengths such as the high distinguishability of the Normal class, while at the same time revealing weaknesses such as low recall or F1-scores for less represented or more complex classes such as Stress and Personality Disorder. This kind of analysis will inform future improvements in data collection, feature engineering, and targeted model changes toward optimizing overall predictive performance.

Table II – Class-Wise Performance

Class	Precision	Recall	F1-Score
Anxiety	0.80	0.79	0.80
Bipolar	0.91	0.66	0.76
Depression	0.69	0.74	0.71
Normal	0.83	0.95	0.89
Personality Disorder	0.96	0.44	0.60
Stress	0.81	0.38	0.52
Suicidal	0.67	0.64	0.66

Normal class gave the highest recall (0.95), denoting an ease to distinguish statements non-pathological. Bipolar and Anxiety classes gave moderate F1 scores (~0.75–0.80), proving that such texts are rich in emotion but are still reliable for identification. Stress and Personality Disorder have

specific problems related to fewer samples and common linguistic expressions, which show much more improvement potential through future data augmentation or feature engineering targeting.

D. ROC-AUC Analysis

The application of ROC-AUC values is thus extended to analyze the discriminative property of an ensemble model for each individual mental health class. In general terms, ROC-AUC indicates how generally different classes can be separated without a threshold, high scores in this measurement meaning that the model cannot efficiently separate the given class from all other classes in the competition. These scores imply that for such cases of extreme imbalance of such classes, one becomes cognizant of how proficiently the classifier steers different lexicons and semantic patterns characterizing each class. Thus, the AUC properly complements traditional true classification metrics such as accuracy, precision, recall, F1 score, by bringing forth ranks of the predictions by the model, so that it becomes possible to allow a finer granularity of class level and clinically interpretable performance.

Table III – One-vs-Rest ROC-AUC Scores

Class	ROC-AUC Score
Anxiety	0.9652
Bipolar	0.9541
Depression	0.9018
Normal	0.9777
Personality Disorder	0.9450
Stress	0.9436
Suicidal	0.9203

An ROC-AUC score above 0.90 implies a very strong class separability and discriminative ability of the ensemble classifier. Highest AUC was true for normal and anxiety classes, which would indicate that more easily detected linguistic patterns were present. Lower AUC values for the depression and suicidal classes suggest that there were certain linguistic cues that overlapped with each other-and this points toward the possibility that further refinement of features may be necessary when distinguishing subtle mental states.

The machine learning comparison includes Logistic Regression, Random Forest, Long Short-Term Memory, Bidirectional LSTM, and an Ensemble of LR + RF models. These models have been compared with accuracy, precision, recall, and F1-score as the major four performance metrics presented in Fig.2. Overall, the ensemble model achieves the best performance, particularly on F1-score, which demonstrates some compromise between precision and recall that is fairly balanced. The BiLSTM does quite well with respect to recall, which indicates great sensitivity in the model in correctly identifying true positives, albeit at the cost of slightly lowering its precision, thus increasing its false-positive rate. Random Forest managed to achieve good precision as well as good accuracy, meaning that it has been effective to reduce false alarms. On the flip side, Logistic Regression performed poorly across the scores, with particularly low Recall, indicating its inability to pick true positive cases. It gives a fair score but is below BiLSTM and the Ensemble model options, giving a clear indication of the added value of modeling sequentially in a bidirectional way.

E. Prediction Output and Confusion Matrix Analysis

The prediction output and confusion matrix provide quite a bit of the performance of the ensemble model on unseen test data. Overall performance measures, on the other hand, indicate to what extent accuracy and F1-score were obtained at the total model level; the confusion matrix shows in detail how particular individual classes have been correctly or not correctly classified. This way, output enters into identifying patterns such as which mental health categories are mixed with others, which well-represented classes the model predicts well, and those that the model poorly predicts, that is, the minor or underlapping classes. In this manner, actionable insights could be provided to the researchers concerning further fine-tuning of the models, alongside feature engineering, and augmenting data to improve classification reliability. Figure 3 shows the output of the model for a sample input sentence: "I want to die." In this case, the classifier predicts the case under the "Suicidal" class with a confidence score of 0.8608. The result is highly indicative of the model's sensitivity to emotionally loaded language, as well as its ability to reveal their signs of suicidal ideation.

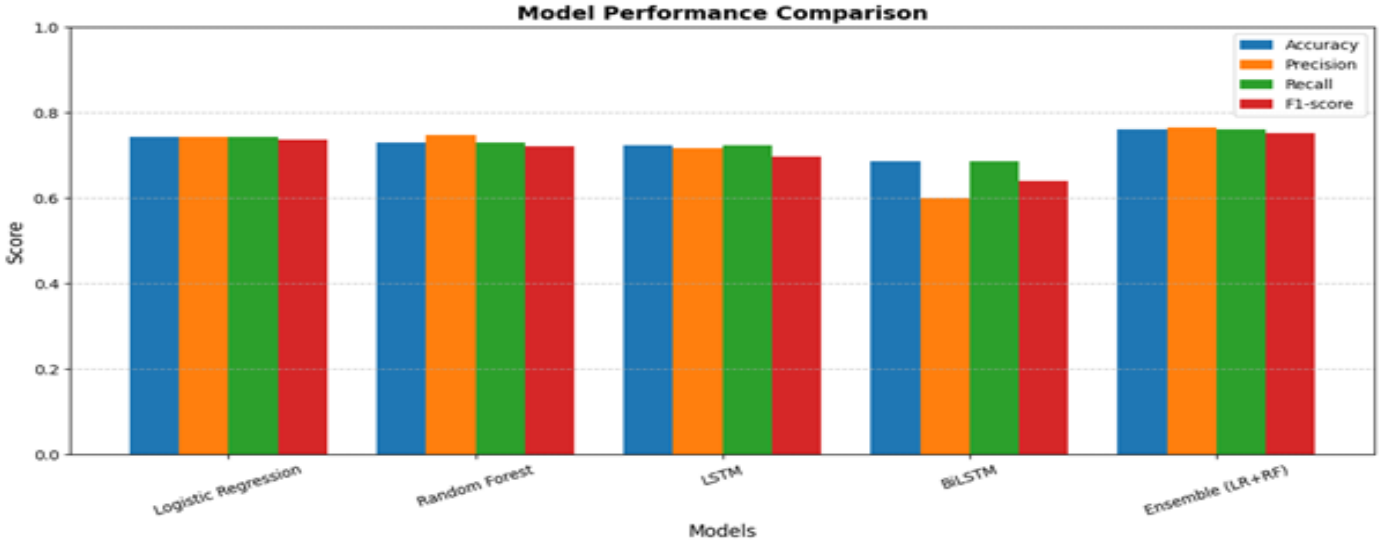
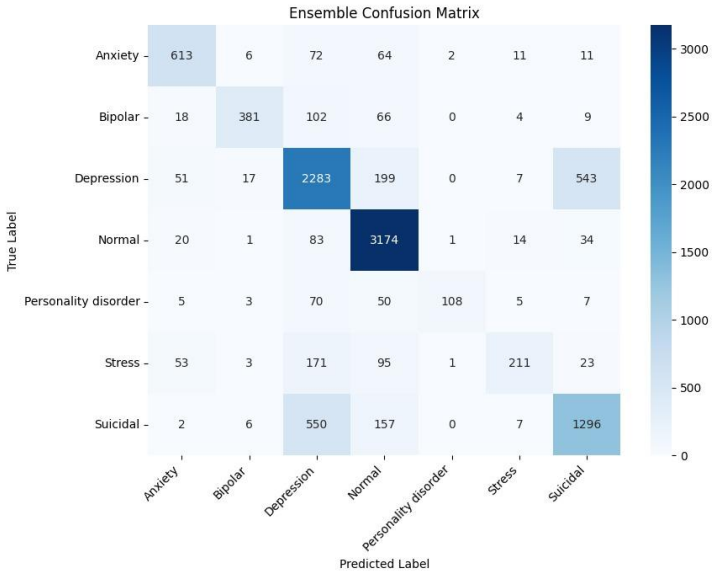


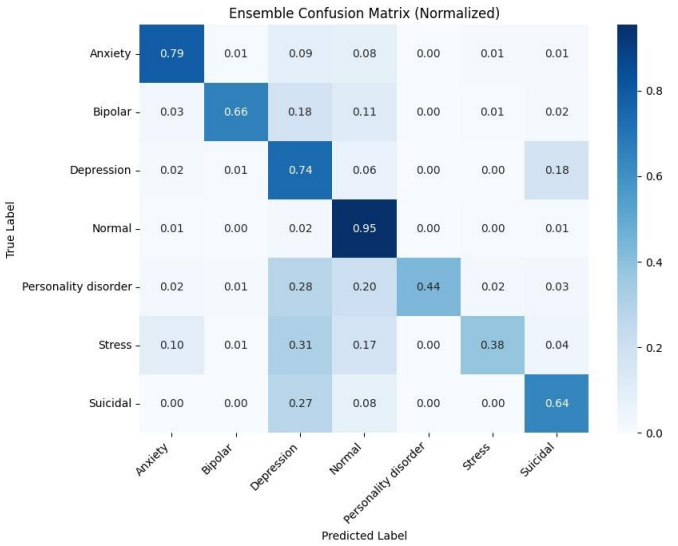
Fig.3. Model Performance Comparison

Fig 4. Confusion Matrix of Ensemble Model



In the ensemble classifier, performance analysis in seven diverse mental health classes: Anxiety, Bipolar, Depression, Normal, Personality Disorder, Stress, and Suicidal are shown in the confusion matrix (Figure 4). Along the diagonal are the correct predictions, and off-diagonal the misclassified ones. Strong classification accuracy is observed for the Normal class (3174 correct predictions) and for Anxiety (613), both having clear linguistic patterns. Misclassifications between Depression and Suicidal classes, however, are much more pronounced (e.g., 543 Suicidal samples predicted as Depression), suggesting some semantic overlap. Stress and Personality Disorder classes appear to have lower recall, indicating that these classes suffer from various challenges arising from class imbalance and overlapping textual features. This matrix informs the strengths and weaknesses of the model in multi-class predictions around mental health.

Fig.5. Normalized Confusion Matrix of Ensemble Model on Test Set



In Fig.5 presents the normalized confusion matrix for the ensemble classifier, showing the proportion of correct and incorrect predictions across seven mental health categories. Each row represents the true class distribution, normalized to sum to 1, allowing for clearer comparison across imbalanced classes. The model demonstrates high accuracy for the Normal class (0.95), indicating strong separability of non-pathological text. Anxiety and Depression also show reliable predictions with normalized scores of 0.79 and 0.74, respectively. In contrast, Personality Disorder and Stress exhibit lower classification accuracy (0.44 and 0.30), with frequent misclassifications into Depression and Suicidal categories. These patterns highlight the model's strengths in detecting well-defined emotional states and its limitations in handling nuanced or overlapping mental health expressions.

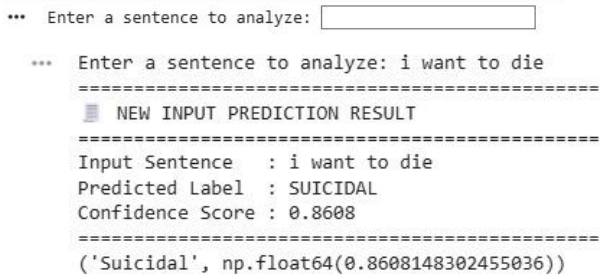


Fig.6. Sentence-Level Prediction Output

F. Feature Importance and Model Interpretability

The interpretability of combined models' predictions in mental health classifications will result in an openness conducive for taking actionable decisions. Feature importance analysis attempted to find out the contribution of TF-IDF, Word2Vec embeddings, MiniBioBERT contextual embeddings, and even sentiment scores in mental illness. TF-IDF flagged lexical cues highly brought considerably influential emotional words to classes such as Suicidal, Anxiety, and Bipolar. Word2Vec embeddings help clarify this semantic context concerning the nuance of meaning vis-a-vis interchangeable keywords like Stress vs. Depression. With MiniBioBERT contextual embeddings, the model can actually capture the meaning of different multiword phrases or even whole sentences. This has immensely increased the ensurement of subtle phrases such as "I feel like ending it." Sentiment features lent this analysis an affect-based side to discriminate between these expressions being positive, neutral, and negative across categories. In a linear setting, Logistic Regression describes the importance of individual features on explainability; Random Forest does so but allows understanding of interactions that may be non-linearly related and of the more combined effect. Therefore, inferentially, the ensemble addresses interpretability itself, that is, these predictions seize no longer appearance; rather, they set out to be trustworthy for clinical decision-making and subsequent informed interventions. In sum, these hybrid feature space and ensemble model types provide a balanced trade-off between predictive performance and explainability, thus qualifying them to be applied in the real world of mental health.

V. Conclusion

This study presents a complete framework of mental health classification using multi-modal textual features; oriented ensemble learning. The novel idea of the method proposed lies in the integration of TF–IDF, Word2Vec semantic embeddings, MiniBioBERT contextual embeddings, and Sentiment features, which capture lexical, semantic, contextual, and affective content of mental health text data. Combining classical ML models (Logistic Regression and Random Forest) into an ensemble classifier helps raise

predictive accuracy and robustness against class imbalances, efficiently capturing subtle patterns among various mental health classes. The ensemble model's performance is much better than its individual counterparts, carrying an accuracy of 76.03% and an F1-score of 0.7531 in its favor, hence affirming the civic nature of the model. Further investigation resulted in stringent performance for classes defined with strong semantic clues such as Normal, Anxiety, and Bipolar, revealing the near-impossibility of detecting classes like Stress and Personality Disorder with impinged or overlapped features. The ROC-AUC analysis contributed towards justifying the good discriminative capacity of the model to support clinical decision-making.

A major innovation of the span of this research is the fine adoption of hybrid feature extraction using classical statistical features, semantic embeddings, contextualized representations, and sentiment analysis. This very high-dimension feature space encourages extrinsic explanation for some degree of interpretability because feature importance analysis throws light on how individual constituents of the features drive predictions. Ensembles are also used due to the simple reason that the results are interpretable; those in the mental health profession can understand the rationale behind the prediction, thereby applying it to a situation like early intervention.

The interpretability of the framework is supported by feature importance and sentence-level prediction analysis; hence its transparency could spur action, bridging the gap between AI models and clinical care. By addressing the class imbalance, leveraging complementary features, and providing explainable outputs, this study proposes-a concept for an integrated methodology towards highly robust, scalable, and clinically relevant automated mental health text classification. Future work can be performed with more extensive datasets into multi-lingual support and advanced transformer-based architecture with an ability to capture the subtlest of linguistic nuances. Furthermore, integration of real-time feedback mechanisms and longitudinal monitoring will help further increase the robustness and usefulness of the proposed tool in the domain of mental health. The study marks the very first step toward implementing more practically applicable and trustworthy AI-assisted solutions for mental health by supporting the idea of using interpretable AI models.

REFERNECES

[1] R. Yulistiani and F. I. Kurniadi, "The Influence of Social Media on Stress Detection Using TF-IDF and Machine Learning Strategies," 2024 International Conference on ICT for Smart Society (ICISS), Bandung, Indonesia, 2024

[2] A. Helmy, R. Nassar and N. Ramadan, "Depression Detection from Social Media Platforms: A Systematic Literature Review," 2023 Intelligent Methods, Systems, and Applications (IMSA), Giza, Egypt, 2023

[3] Shahid Munir Shah, Mahmoud Mohammad Aljawarneh, Muhammad Aamer Saleem & Mahmoud Saleh Jawarneh. (2024). Mental illness detection through harvesting social media: a comprehensive literature review. PeerJ Computer Science, 2024.

[4] Helmy, A. M., Nassar, R., & Ramadan, N. (2024). Depression detection for Twitter users using sentiment analysis in English and Arabic tweets. Artificial Intelligence in Medicine, 147, Article 102716.

[5] M. R. Ningsih and J. Unjung, "Sentiment analysis on social media using TF-IDF vectorization and H2O gradient boosting for student anxiety detection," Scientific Journal of Informatics, vol. 11, no. 4, pp. 1137–1144, Nov. 2024.

[6] Y. Shen, H. Yang and L. Lin, "Automatic Depression Detection: an Emotional Audio-Textual Corpus and A Gru/Bilstm-Based Model," ICASSP 2022 - 2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Singapore, Singapore, 2022

[7] G. Lam, H. Dongyan and W. Lin, "Context-aware Deep Learning for Multi-modal Depression Detection," ICASSP 2019 - 2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Brighton, UK, 2019

[8] D. Pakkattil and R. S. Devi, "Empowering Mental Health: CNN and LSTM Fusion for Timely Depression Detection in Women," Int. J. Electr. Comput. Eng. Syst., vol. 15, no. 8, pp. 631–640, Sep. 2024

[9] Chen, Z., Yang, R., Fu, S., Zong, N., Liu, H., & Huang, M. (2024, January 29). Detecting Reddit users with depression using a hybrid neural network SBERT-CNN (Version 2)

[10] N. K. Iyortsuun, S.-H. Kim, M. Jhon, H.-J. Yang, and S. Pant, "A Review of Machine Learning and Deep Learning Approaches on Mental Health Diagnosis," *Healthcare*, vol. 11, no. 3, Art. 285, 2023.

[11] A. Mohamed, R. AlAleeli, and K. Shaalan, "Advancing Predictive Healthcare: A Systematic Review of Transformer Models in Electronic Health Records," *Computers*, vol. 14, no. 4, Art. 148, Apr. 2025.

[12] B. G. Bokolo and Q. Liu, "Advanced Comparative Analysis of Machine Learning and Transformer Models for Depression and Suicide Detection in Social Media Texts," *Electronics*, vol. 13, no. 20, Art. 3980, Oct. 2024

[13] Chung, MK., Lee, S.Y., Shin, T. et al. BERT and BERTopic for screening clinical depression on open-ended text messages collected through a mobile application from older adults. BMC Public Health 25, 2161 (2025).

[14] Hussain, N.; Qasim, A.; Mehak, G.; Zain, M.; Sidorov, G.; Gelbukh, A.; Kolesnikova, O. Multi-Level Depression Severity Detection with Deep Transformers and Enhanced Machine Learning Techniques. AI 2025

[15] Narvaez Burbano, R.; Caicedo Rendon, O.M.; Astudillo, C.A. An Encoder-Only Transformer Model for Depression Detection from Social Network Data: The DEENT Approach. Appl. Sci. 2025.

[16] A. Malhotra and R. Jindal, "XAI Transformer based Approach for Interpreting Depressed and Suicidal User Behavior on Online Social Networks," *Cogn. Syst. Res.*, vol. 84, p. 101186, 2024.

[17] E. Kerz, S. Zanwar, Y. Qiao, and D. Wiechmann, "Toward explainable AI (XAI) for mental health detection based on language behavior," *Frontiers in Psychiatry*, vol. 14, Art. 1219479, Dec. 2023.

[18] Hulsen, T. Explainable Artificial Intelligence (XAI): Concepts and Challenges in Healthcare. AI 2023, 4, 652-666

[19] Sheu, R.-K.; Pardeshi, M.S. A Survey on Medical Explainable AI (XAI): Recent Progress, Explainability Approach, Human Interaction and Scoring System. Sensors 2022, 22, 8068.

[20] Hildt, E. What Is the Role of Explainability in Medical Artificial Intelligence? A Case-Based Approach. Bioengineering 2025, 12, 375